

way to store energy without converting DC into AC.

A thermopower device made of carbon nanotubes could output the same energy as a lithium-ion battery but at 1/100th the size.

Li-ion batteries get nanocomposite boost

Researchers at Georgia Institute of Technology have found that hanging silicon particles of nanometre size on the tress of carbon black can boost the capacity of lithium-ion batteries fivefold. These self-assemble into porous spheres that increase an electrode's surface area with interconnected internal channels. The new hybrid silicon-carbon nano-electrodes can be massproduced economically.

At present, high-performance lithium-ion batteries use anodes made from carbon (graphite). Silicon, as a substitute for graphite, offers a theoretical improvement of ten times in capacity over graphite.

Cotton threads to charge portable gadgets

Juan Hinestroza, assistant professor in Fiber Science and Apparel Design, Cornell University, USA, has developed a cotton thread that can conduct electric current as well as a metal wire can.

The technology allows cotton to remain flexible, light and comfortable while being electronically conductive. Previous technologies have achieved conductivity but the resulting fibre becomes rigid and heavy. The new technique makes yarns friendly to further processing such as weaving, sewing and knitting. It is unique in that simple

knots in such specially treated threads can complete a circuit and power electronic devices such as iPods.

Shake your mobile to recharge it

Now all you need to do to recharge your cell phone is to shake it a bit. Nokia would soon launch such a phone that recharges itself by using the energy from the user's motion.

In the phone, the heavier components, such as the radio transmitter circuit and battery, are supported on a sturdy frame. This frame can move along two sets of rails: one allows it to travel up and down, while the other allows it to move from side to side.

Strips of piezoelectric crystals at the end of each rail generate a current when compressed by the frame. So as the user walks, or moves the phone, the motion generates electricity. This

charges a capacitor, which, in turn, trickles charge into the battery, keeping it topped up.

Smart windows to save energy

California-based Sage Electrochromics has developed smart windows to save energy. The technology uses a series of ceramic material layers deposited onto sheets of glass to form an electrochromic device that allows the user to switch the window between a clear state and a highly tinted state using a low-voltage electrical current. The windows can be controlled manually or by a building automation system.

According to John Van Dine, founder and chief executive, Sage, this technology can control both visible and near infrared light using the inorganic thin-film coatings that are durable and cost little.



SageGlass windows, skylights and curtainwalls control daylight elegantly and efficiently while maintaining your view and connection to the outdoors

